Tathagat Agrawal

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Applied ML researcher and scientist building end-to-end production AI systems: LLM fine-tuning, RAG, agentic tool-calling, evaluation frameworks, multi-task NER and learning-to-rank. Published 3 peer-reviewed papers.

Technical Skills

Languages and tools: Python, C/C++, Java, JavaScript, R; Git, GitHub, Claude Code, Cursor, LaTeX
Generative AI and LLMs: LLM fine-tuning (SFT, LoRA), Retrieval-Augmented Generation (RAG), embeddings and vector search (ChromaDB, Sentence-Transformers), prompt engineering, agentic AI and tool calling (LangGraph, LangChain, ReAct), LLM evaluation, hallucination mitigation, OpenAI and Gemini APIs, LLaMA-Factory
ML and Deep Learning: PyTorch, TensorFlow, Keras, Scikit-Learn, XGBoost, HuggingFace Transformers, learning-to-rank, NER, SHAP, OpenCV, spaCy, NLTK
Deployment and MLOps: FastAPI, Flask, REST APIs, Pydantic validation, MLflow, Azure ML, model monitoring
Data and Visualisation: Pandas, NumPy, data pipelines and feature engineering, Matplotlib, Seaborn, Power BI, Excel

Work Experience

Applied Scientist (AI/ML) — **October 2025 – Present**
Neural Nurture — *Indore, MP, India*

- Built a **learning-to-rank recommendation system** (LambdaMART/XGBoost) for the travel industry with custom relevance features, **improving NDCG@3 by about 40%** over the prior baseline; added SHAP/attention-weight explainability for end users.
- **Fine-tuned LLMs** using SFT and LoRA to generate natural language explanations for recommendations; diagnosed and resolved a training data bias causing hallucinations, **improving factual accuracy to 98.4%**.
- Built and trained a **multi-task NER + intent classification model** (RoBERTa backbone) on a custom dataset of 100,000 annotated samples covering 44 entity classes, achieving an F1-score of over 94%.
- Designed the **agentic tool-calling architecture** for a conversational travel application, including prompt engineering and external API integration (ToS compliance, cost optimization), to route structured NER output to the right tools.
- Designed an **end-to-end evaluation framework**, leading to early bug discovery and continuous performance monitoring.
- Translated recent **ML research** into production-ready prototypes through controlled experiments across LLMs and conventional ML models; presented experimental results and architectural proposals to senior leadership, influencing product roadmap decisions; conducted technical interviews for ML internship candidates.

Projects

Anuchhed: citation-first Q&A over the Constitution of India — **November 2026**
Python, LangGraph, FastAPI, Gemini API, Qwen3 embeddings, SQLite FTS5 — *GitHub*

- Built **citation-first Q&A** over the Constitution of India: the official PDF parsed into a provision tree with amendment history, and **hybrid retrieval** (reference lookup, BM25, dense embeddings, cross-encoder reranking) chosen through an eight-configuration ablation.
- Every statement cites its provision and every quote is **verified against the official text**; on a 70-question development set, 97% correct behaviour versus 87% answering from memory and 71% for naive RAG, with 100% exact quotes.

Agentic Eval Framework — **September 2026**
Python, Gemini API, LangGraph, LangChain, SQLite, Pydantic — *GitHub*

- Built an architecture-agnostic framework scoring agentic LLM systems on six deterministic, rule-based metrics (numeric correctness, dual groundedness checks, contextual/clarification/refusal correctness, tool-call efficiency) with no LLM-as-judge.
- Implemented four interchangeable agent architectures (LangGraph state machine, ReAct, context-stuffing, plan-and-execute) behind one adapter interface; the comparison report shows concrete trade-offs (e.g., 50%-vs-100% groundedness).

AI Clinic Scheduler — **August 2026**
LangGraph, LangChain, FastAPI, Pydantic, ChromaDB, Sentence-Transformers, OpenRouter — *GitHub*

- Built an LLM-powered scheduling agent (LangGraph ReAct) that interprets patient symptoms and recommends a doctor via RAG over specialties and bios (Sentence-Transformers embeddings, Chroma vector store, callable retriever tool).
- Defined Pydantic schemas for every tool call so malformed arguments surface as correctable errors; built a FastAPI backend (chat, doctor/slot lookups, admin) and a no-build-step HTML/JS frontend for booking and receptionist admin.

Other projects:

- **Trade Filter ML** (Python, LightGBM, scikit-learn, Optuna, MLflow, SHAP, pandas, November 2026): leakage-safe meta-labeling study with a pre-registered holdout and an honestly reported null result
- **Multilingual NER with mBERT** (PyTorch, HuggingFace, Fine-tuning, NLP, December 2023): mBERT fine-tuned for Hindi and English NER (F1 0.899 and 0.946)

Publications

[1] Agrawal T, Kumar M (2025) Grokking in Neural Networks: A Review. SN Computer Science 6:627. DOI link

[2] Agrawal T, Kumar M (2025) Analysis of Grokking-Induced Rigidity in Neural Networks using Transfer Learning. In: Proceedings of Data Analytics and Management, ICDAM.

[3] Agrawal T, Chauhan A, Singh A (2024) Advancing Linguistic Frontiers with mBERT Fine-Tuning for Hindi and English NER Using HiNER and WikiNEuRal Datasets. In: ICCCNT 2024. Paper link. F1 of 0.899 (HiNER), 0.946 (WikiNEuRal).

Education

Integrated M. Tech. in AI — **2020 – 2025**
VIT Bhopal University, Bhopal — *CGPA: 9.1/10.0*

- **Master's thesis:** *Investigating the Effects of Grokking Phenomenon in Transfer Learning Settings* (publications [1], [2]).